```{r}
# 2.22(a)
df1 <- star5_small[star5_small$aide == 0, ]
lm_a <- lm(totalscore ~ small, data = df1)
summary(lm_a)
```

```
Call:
lm(formula = totalscore ~ small, data = df1)

Residuals:
    Min      1Q  Median      3Q     Max 
-189.44  -50.44   -9.44   43.06  324.38 

Coefficients:
            Estimate Std. Error t value Pr(>|t|)    
(Intercept)  916.442      3.675 249.396   <2e-16 ***
small         12.175      5.369   2.268   0.0236 *  
---
Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1

Residual standard error: 74.59 on 773 degrees of freedom
Multiple R-squared:  0.006608,	Adjusted R-squared:  0.005323 
F-statistic: 5.142 on 1 and 773 DF,  p-value: 0.02363
```

```{r}
# 2.22(b)
lm_b1 <- lm(readscore ~ small, data = df1)
summary(lm_b1)
lm_b2 <- lm(mathscore ~ small, data = df1)
summary(lm_b2)
```

```
Call:
lm(formula = readscore ~ small, data = df1)

Residuals:
    Min      1Q  Median      3Q     Max 
-74.665 -21.590  -2.665  16.410 194.335 

Coefficients:
            Estimate Std. Error t value Pr(>|t|)    
(Intercept)  432.665      1.488 290.760  < 2e-16 ***
small          6.924      2.174   3.185  0.00151 ** 
---
Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1

Residual standard error: 30.2 on 773 degrees of freedom
Multiple R-squared:  0.01295,	Adjusted R-squared:  0.01167 
F-statistic: 10.14 on 1 and 773 DF,  p-value: 0.001507
```

```
Call:
lm(formula = mathscore ~ small, data = df1)

Residuals:
     Min       1Q   Median       3Q      Max 
-144.777  -35.028   -5.028   29.223  142.223 

Coefficients:
            Estimate Std. Error t value Pr(>|t|)    
(Intercept)  483.777      2.449 197.545   <2e-16 ***
small          5.251      3.578   1.467    0.143    
---
Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1

Residual standard error: 49.71 on 773 degrees of freedom
Multiple R-squared:  0.002778,	Adjusted R-squared:  0.001488 
F-statistic: 2.153 on 1 and 773 DF,  p-value: 0.1427
```

```{r}
# 2.22(c)
df2 <- star5_small[star5_small$small == 0, ]
lm_c <- lm(totalscore ~ aide, data = df2)
summary(lm_c)
```

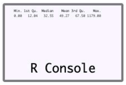
```
Call:
lm(formula = totalscore ~ aide, data = df2)

Residuals:
     Min       1Q   Median       3Q      Max 
-191.748  -50.442   -5.442   40.558  312.558 

Coefficients:
            Estimate Std. Error t value Pr(>|t|)    
(Intercept)  916.442      3.559 257.529   <2e-16 ***
aide           4.306      4.994   0.862    0.389    
---
Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1

Residual standard error: 72.23 on 835 degrees of freedom
Multiple R-squared:  0.0008898,	Adjusted R-squared:  -0.0003068 
F-statistic: 0.7436 on 1 and 835 DF,  p-value: 0.3888
```

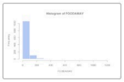
```{r}
# 2.22(d)
lm_d1 <- lm(readscore ~ aide, data = df2)
summary(lm_d1)
lm_d2 <- lm(mathscore ~ aide, data = df2)
summary(lm_d2)
```

```
Call:
lm(formula = readscore ~ aide, data = df2)

Residuals:
    Min      1Q  Median      3Q     Max 
-74.665 -21.536  -2.536  15.464 194.335 

Coefficients:
            Estimate Std. Error t value Pr(>|t|)    
(Intercept)  432.665      1.460 296.341   <2e-16 ***
aide           2.871      2.049   1.401    0.161    
---
Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1

Residual standard error: 29.64 on 835 degrees of freedom
Multiple R-squared:  0.002347,	Adjusted R-squared:  0.001152 
F-statistic: 1.964 on 1 and 835 DF,  p-value: 0.1615
```

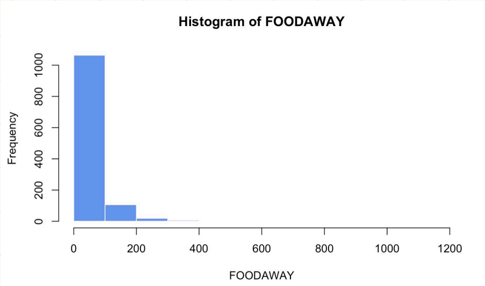
```
Call:
lm(formula = mathscore ~ aide, data = df2)

Residuals:
     Min       1Q   Median       3Q      Max 
-146.212  -31.212   -5.777   29.223  142.223 

Coefficients:
            Estimate Std. Error t value Pr(>|t|)    
(Intercept)  483.777      2.355 205.464   <2e-16 ***
aide           1.435      3.304   0.434    0.664    
---
Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1

Residual standard error: 47.79 on 835 degrees of freedom
Multiple R-squared:  0.0002258,	Adjusted R-squared:  -0.0009715 
F-statistic: 0.1886 on 1 and 835 DF,  p-value: 0.6642
```

```{r}
# 2.25(a)
summary(cex5_small$foodaway)
hist(cex5_small$foodaway, main = "Histogram of FOODAWAY",
     xlab = "FOODAWAY", col = "cornflowerblue", border = "white")
```





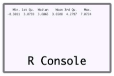
```
   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
   0.00   12.04   32.55   49.27   67.50 1179.00
```

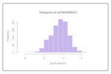



```{r}
# 2.25(b)
summary(cex5_small$foodaway[cex5_small$advanced == 1])
summary(cex5_small$foodaway[cex5_small$college == 1])
summary(cex5_small$foodaway[cex5_small$advanced == 0 & cex5_small$college == 0])
```

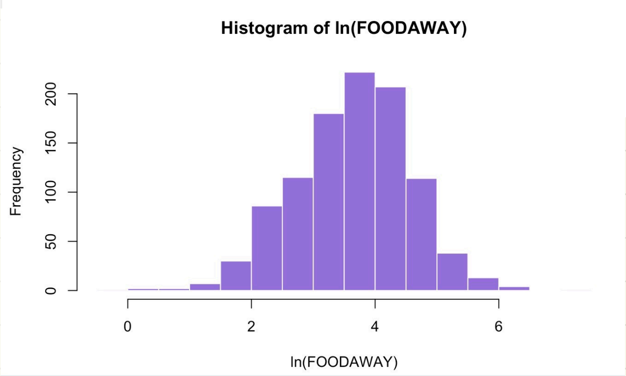
```
   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
   0.00   21.67   48.15   73.15   90.00 1179.00
   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
   0.00   14.44   36.11   48.60   68.67  416.11
   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
   0.00    9.63   26.02   39.01   52.65  437.78
```

```{r}
# 2.25(c)
pos <- cex5_small$foodaway > 0
summary(log(cex5_small$foodaway[pos]))
hist(log(cex5_small$foodaway[pos]),
     main = "Histogram of ln(FOODAWAY)",
     xlab = "ln(FOODAWAY)", col = "mediumpurple", border = "white")
```





```
   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
-0.3011  3.0759  3.6865  3.6508  4.2797  7.0724
```




```{r}
# 2.25(d)
cex_d <- cex5_small[cex5_small$foodaway > 0, ]
lm_e <- lm(log(foodaway) ~ income, data = cex_d)
summary(lm_e)
```

```
Call:
lm(formula = log(foodaway) ~ income, data = cex_d)

Residuals:
    Min      1Q  Median      3Q     Max 
-3.6547 -0.5777  0.0530  0.5937  2.7000 

Coefficients:
             Estimate Std. Error t value Pr(>|t|)    
(Intercept) 3.1293004  0.0565503   55.34   <2e-16 ***
income      0.0069017  0.0006546   10.54   <2e-16 ***
---
Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1

Residual standard error: 0.8761 on 1020 degrees of freedom
Multiple R-squared:  0.09826,	Adjusted R-squared:  0.09738 
F-statistic: 111.1 on 1 and 1020 DF,  p-value: < 2.2e-16
```

```
# 2.25(e)
plot(cex_d$income, log(cex_d$foodaway),
     main = "ln(FOODAWAY) vs INCOME",
     xlab = "Income ($100 units)",
     ylab = "ln(FOODAWAY)",
     pch = 1, col = "dodgerblue")
abline(lm_e, col = "tomato", lwd = 2)
```